Meeting Memo

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>
> Content: Explain Inventory(or Stock) and Ranking

Inventory

Basic Formula

```
期初库存 + 进货数量 - 期末库存 = 实际销量

或者

期初库存 + 进货数量 - 实际销量 = 期末库存
```

Calculate Beginning of Stock

```
期初库存 =
CALCULATE(
    SUM('库存表'[库存数量]),
    FILTER(
        ALL('库存表'[日期]),
        '库存表'[日期] = STARTOFMONTH('库存表'[日期])
    )
)
```

Calculate Ending of Stock

```
期末库存 =
CALCULATE(
    SUM('库存表'[库存数量]),
    FILTER(
        ALL('库存表'[日期]),
        '库存表'[日期] = ENDOFMONTH('库存表'[日期])
    )
)
```

Calculate Stock Summary Table

```
库存汇总表 =
ADDCOLUMNS(
    SUMMARIZE(
        '库存表',
        '库存表'[商品],
        '库存表'[日期].[年份],
        '库存表'[日期].[月份],
        "库存月份", FORMAT('库存表'[日期], "yyyy年MM月")
    ),
    "期初库存", [期初库存],
    "期末库存", [期末库存]
)
```

What if the start date and end date is not the actual and end date

> 当库存表中没有完整的每日数据时，我们需要先找到每个商品在每个月实际存在的最小日期（当月最早记录日期）和最大日期（当月最晚记录日期），再基于这些实际日期计算库存。可以使用以下 DAX 方案：

Calculate Minimum Actual Date

```
当月实际最小日期 =
CALCULATE(
    MIN('库存表'[日期]),
    ALLEXCEPT('库存表', '库存表'[商品], '库存表'[日期].[年份], '库存表'[日期].[月份])
)
```

Calculate Maximum Actual Date

```
当月实际最大日期 =
CALCULATE(
    MAX('库存表'[日期]),
    ALLEXCEPT('库存表', '库存表'[商品], '库存表'[日期].[年份], '库存表'[日期].[月份])
)
```

Calculate Acutal Beginning of Stock

```
实际期初库存 =
VAR CurrentMinDate = [当月实际最小日期]
RETURN
CALCULATE(
    SUM('库存表'[库存数量]),
    FILTER(
        ALLEXCEPT('库存表', '库存表'[商品], '库存表'[日期].[年份], '库存表'[日
```

```
期].[月份]),
        '库存表'[日期] = CurrentMinDate
    )
)
```

Calculate Acutal Ending of Stock

```
实际期末库存 =
VAR CurrentMaxDate = [当月实际最大日期]
RETURN
CALCULATE(
    SUM('库存表'[库存数量]),
    FILTER(
        ALLEXCEPT('库存表', '库存表'[商品], '库存表'[日期].[年份], '库存表'[日
期].[月份]),
        '库存表'[日期] = CurrentMaxDate
    )
)
```

Calculate Stock Summary Table

```
实际库存汇总表 =
FILTER(
    ADDCOLUMNS(
        SUMMARIZE(
            '库存表',
            '库存表'[商品],
            '库存表'[日期].[年份],
            '库存表'[日期].[月份],
            "库存月份", FORMAT('库存表'[日期], "yyyy年MM月")
        ),
        "期初库存", [实际期初库存],
        "期末库存", [实际期末库存]
    ),
    NOT ISBLANK([期初库存]) && NOT ISBLANK([期末库存])
)
```

Ranking in Power BI

> All the sales data is just a somple

RANKX, RANK Examples

Basic Ranking using RANKX

```
Sales_Rankx_1 =
IF(

    HASONEVALUE(Dim_Store[BTQ]),  // use HASONEVALUE to evaluate if the current iteration is within the scope of a single store row

    RANKX(ALLSELECTED(Dim_Store[Region],Dim_Store[BTQ]), [Sales],, DESC, Skip),

    BLANK() // return BLANK value if not a single store row
)
```

```
Sales_Rankx_2 =
IF(
    HASONEVALUE(Dim_Store[BTQ]),  // use HASONEVALUE to evaluate if the current iteration is within the scope of a single store row

    RANKX(ALL(Dim_Store[Region],Dim_Store[BTQ]), [Sales],, DESC, Skip), // order sales by all store

    BLANK() // return BLANK value if not a single store row

)
// when using ALL function, the visual will not be affected by any slicer
```

> Difference from above expression, is `ALLSELECTED` and `ALL`

Basic Ranking using RANK

```
Sales_Rank_1 =

RANK(DENSE,

ALLSELECTED(Dim_Store[Region], Dim_Store[BTQ]),

ORDERBY([Sales], DESC, Dim_Store[Region], ASC, Dim_Store[BTQ], ASC)
)
```

RANKX, RANK Difference

- `RANKX` is earlier than `RANK` function, `RANK` function is a simplified version of `RANKX`.
- `RANKX` will return 1 for "Total" in table visual, need to use `HASONEVALUE` function for iteration evaluations, in order to elimiate the 1 value.
- In `RANK` function, developers must explicitly specify the sort order using `ORDERBY` function. and `RANK` support mutiple column orders,

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Reminder

> Make sure use `ALLSELECTED` function instead of `ALL` if you want the ranking result should be affected by the slicers (The context).